DSA06 DATA HANDLING AND VISUALIZATION

Activity: Map Storytelling Challenge | Slot D | Story Brief MS-D12: Illiteracy (% of population, 1970)

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Question Paper

DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot D

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-D12: Illiteracy (% of population, 1970) [20 marks]

Brief: You are building a map story about Illiteracy (% of population, 1970) across the United States for a policy brief for a minister. Your comparison variable is Murder rate (per 100,000, 1976). Your focus region is the North Central region. Your map uses 5 classes and the "mako" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$illit      # the variable you map
W <- d$murder      # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of illit: ggplot(mapdat, aes(long, lat, group = group, fill = illit)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "mako") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 5 classes: equal interval, cut(V, 5), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of illiterate people: count <- d$pop * 1000 * d$illit / 100. Name the top 3 states by count and the top 3 states by illit [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the North Central region. Report the mean of illit in North Central and its ratio to the mean of all 50 states [1]; the highest state in North Central with its value [0.5]; and how many North Central states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a policy brief for a minister [1]. [4]

Set-up (given code, not edited)

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$illit      # the variable you map
W <- d$murder     # the comparison variable
```

Output: data used (V = illit, W = murder, pop in thousands)

state	zone	pop	illit	murder
Alabama	South	3615	2.1	15.1
Alaska	West	365	1.5	11.3
Arizona	West	2212	1.8	7.8
Arkansas	South	2110	1.9	10.1
California	West	21198	1.1	10.3
Colorado	West	2541	0.7	6.8
Connecticut	Northeast	3100	1.1	3.1
Delaware	South	579	0.9	6.2
Florida	South	8277	1.3	10.7
Georgia	South	4931	2.0	13.9
Hawaii	West	868	1.9	6.2
Idaho	West	813	0.6	5.3
Illinois	North Central	11197	0.9	10.3
Indiana	North Central	5313	0.7	7.1
Iowa	North Central	2861	0.5	2.3
Kansas	North Central	2280	0.6	4.5
Kentucky	South	3387	1.6	10.6
Louisiana	South	3806	2.8	13.2
Maine	Northeast	1058	0.7	2.7
Maryland	South	4122	0.9	8.5
Massachusetts	Northeast	5814	1.1	3.3
Michigan	North Central	9111	0.9	11.1
Minnesota	North Central	3921	0.6	2.3
Mississippi	South	2341	2.4	12.5
Missouri	North Central	4767	0.8	9.3

state	zone	pop	illit	murder
Montana	West	746	0.6	5.0
Nebraska	North Central	1544	0.6	2.9
Nevada	West	590	0.5	11.5
New Hampshire	Northeast	812	0.7	3.3
New Jersey	Northeast	7333	1.1	5.2
New Mexico	West	1144	2.2	9.7
New York	Northeast	18076	1.4	10.9
North Carolina	South	5441	1.8	11.1
North Dakota	North Central	637	0.8	1.4
Ohio	North Central	10735	0.8	7.4
Oklahoma	South	2715	1.1	6.4
Oregon	West	2284	0.6	4.2
Pennsylvania	Northeast	11860	1.0	6.1
Rhode Island	Northeast	931	1.3	2.4
South Carolina	South	2816	2.3	11.6
South Dakota	North Central	681	0.5	1.7
Tennessee	South	4173	1.7	11.0
Texas	South	12237	2.2	12.2
Utah	West	1203	0.6	4.5
Vermont	Northeast	472	0.6	5.5
Virginia	South	4981	1.4	9.5
Washington	West	3559	0.6	4.3
West Virginia	South	1799	1.4	6.7
Wisconsin	North Central	4589	0.7	3.0
Wyoming	West	376	0.6	6.9

Task 1. The base map [4 marks]

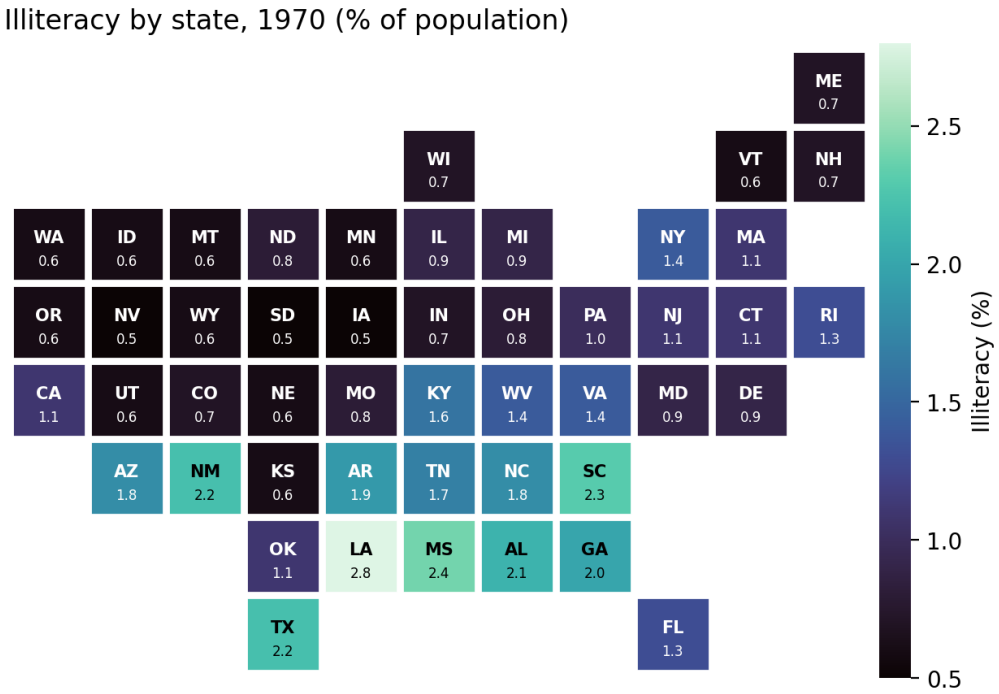
R code

```
ggplot(mapdat, aes(long, lat, group = group, fill = illit)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "mako", name = "Illiteracy (%)") +
  labs(title = "Illiteracy by state, 1970 (% of population)") +
  theme_void()

d[which.max(d$illit), c("state", "illit")]
min(d$illit)
d$state[d$illit == min(d$illit)]
median(d$illit)
```

Output

`Louisiana 2.8` `[1] 0.5` (Iowa, Nevada, South Dakota) `[1] 0.95` (median)



Answer:
Highest state: Louisiana, 2.8%.
Lowest value: 0.5%, shared by three states: Iowa, Nevada and South Dakota.
Median of all 50 states = 0.95%.

Task 2. The regional story [4 marks]

R code

```
zone_mean <- tapply(V, d$zone, mean)
round(zone_mean, 2)
names(which.max(zone_mean)); max(zone_mean)
names(which.min(zone_mean)); min(zone_mean)
max(zone_mean) - min(zone_mean)
cor(V, W)
```

Output

North Central	Northeast	South	West
0.70	1.00	1.74	1.02

Gap (South - North Central): `[1] 1.0375` `cor(V, W) = [1] 0.703`

Answer:
Mean illiteracy by region: North Central = 0.70, Northeast = 1.00, South = 1.74, West = 1.02.
Highest region: South (1.74). Lowest region: North Central (0.70). Gap = 1.74 - 0.70 = 1.04 percentage points.
cor(illit, murder) = 0.70. This is a strong (just above 0.7) positive correlation: states with higher illiteracy also tend to have a higher murder rate. But this is only a correlation, it does not prove that illiteracy causes murder (other things like poverty can affect both).

Task 3. Classes change the story [4 marks]

R code

```
d$eq <- cut(V, 5)
d$qt <- cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]

ggplot(mapdat, aes(long, lat, group = group, fill = eq)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "Illiteracy (%)") +
  labs(title = "Equal interval: cut(V, 5)") + theme_void()
ggplot(mapdat, aes(long, lat, group = group, fill = qt)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "Illiteracy (%)") +
  labs(title = "Quantile classes") + theme_void()

levels(d$eq); table(d$eq)
levels(d$qt); table(d$qt)
```

Output

Class 1	Class 2	Class 3	Class 4	Class 5
(0.498,0.96]	(0.96,1.42]	(1.42,1.88]	(1.88,2.34]	(2.34,2.8]
25	11	5	7	2

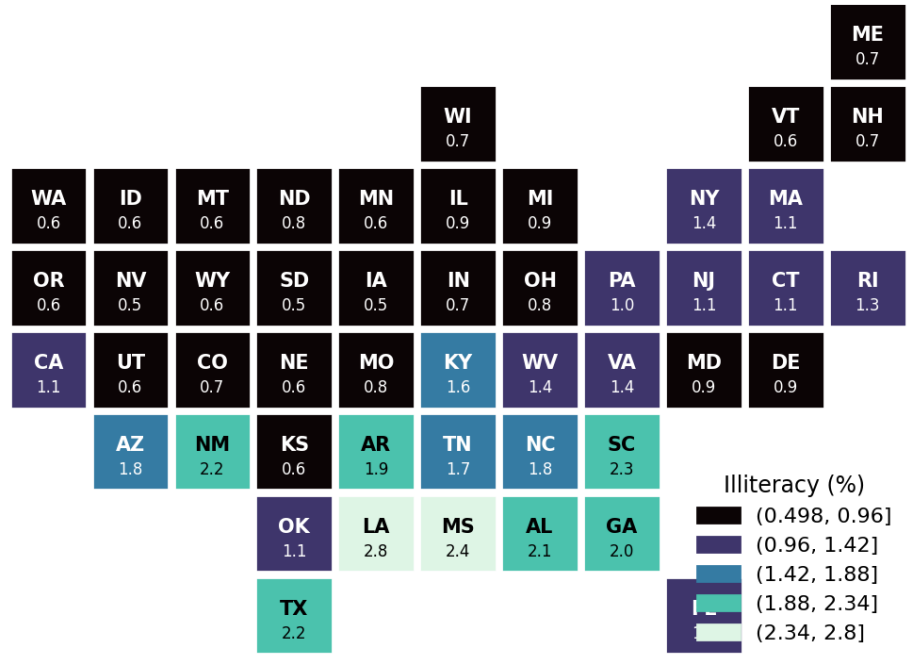
table(d$eq): equal interval (number of states in each class)

Class 1	Class 2	Class 3	Class 4	Class 5
[0.5,0.6]	(0.6,0.8]	(0.8,1.1]	(1.1,1.8]	(1.8,2.8]
13	8	10	10	9

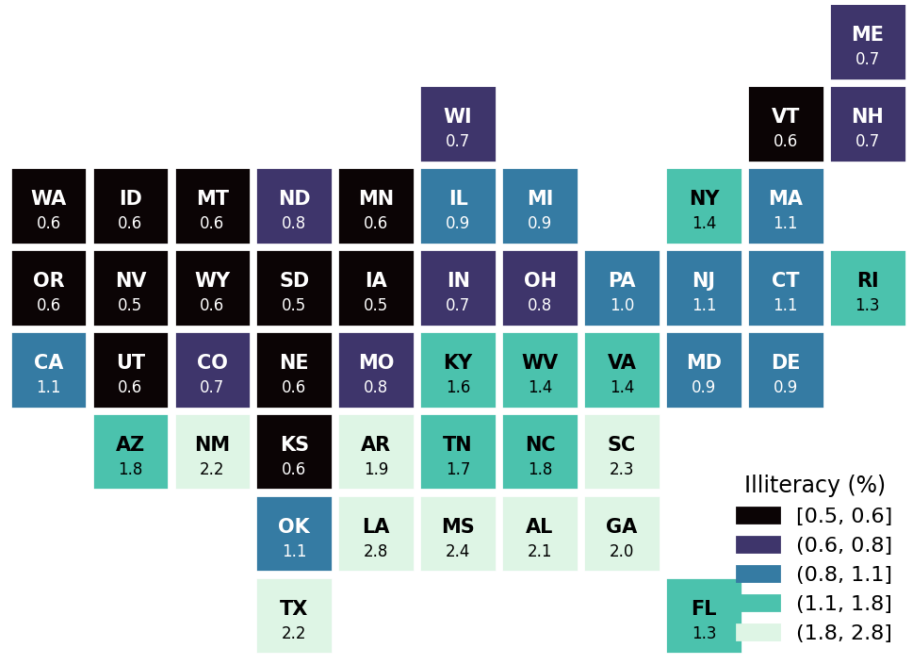
table(d$qt): quantile (number of states in each class)

Task 3 (continued): output maps

Equal interval: cut(V, 5)



Quantile: cut(V, quantile(V, seq(0, 1, length.out = 6)))



Answer:

Equal interval: lower limit of the top class = 2.34, with 2 states in it (Louisiana and Mississippi).

Quantile: lower limit of the top class = 1.8, with 9 states in it (Alabama, Arkansas, Georgia, Hawaii, Louisiana, Mississippi, New Mexico, South Carolina, Texas). Hawaii is counted here but it is not drawn on the map.

Equal interval makes the highest values look rarer, because only 2 states (4%) fall in the top class, while 25 states are in the lowest class.

For a minister I would choose the quantile map. Illiteracy is skewed (most states are low, a few are very high), so equal interval puts half the country in one colour and hides the differences. Quantile classes have about 10 states each, so the minister can easily see which states are in the worst group.

Task 4. Counts versus rates [4 marks]

R code

```
d$count <- d$pop * 1000 * d$illit / 100
top_count <- d$state[order(-d$count)][1:3]
top_rate  <- d$state[order(-d$illit)][1:3]
d[order(-d$count), c("state", "count")][1:3, ]
d[order(-d$illit), c("state", "illit")][1:3, ]
length(intersect(top_count, top_rate))
```

Output

Top 3 by count	count (people)	illit (%)
Texas	269,214	2.2
New York	253,064	1.4
California	233,178	1.1

Top 3 by illit	illit (%)	count (people)
Louisiana	2.8	106,568
Mississippi	2.4	56,184
South Carolina	2.3	64,768

`[1] 0` (states in both lists)

Answer:
Top 3 by count: Texas (269,214), New York (253,064), California (233,178).
Top 3 by illit: Louisiana (2.8), Mississippi (2.4), South Carolina (2.3).
Number of states in both lists = 0.
A map of counts would mislead because the count mostly follows population size, so big states like Texas, New York and California would look worst even though their rates are not high. We should map the rate (percentage of people who are illiterate) instead, so states of different sizes can be compared fairly.

Task 5. The story for the North Central region [4 marks]

R code

```
nc <- d[d$zone == "North Central", ]
mean(nc$illit)
mean(nc$illit) / mean(V)
nc[order(-nc$illit), c("state", "illit")][1:3, ]
sum(nc$illit > median(V))
```

Output

Mean illit (North Central)	Mean illit (all 50)	Ratio	States above median (0.95)
0.70	1.17	0.60	0

Highest in North Central	illit (%)
Illinois	0.9
Michigan	0.9
Missouri / Ohio / North Dakota	0.8

Answer:
Mean illit of North Central = 0.70%. Mean of all 50 states = 1.17%, so the ratio = 0.70 / 1.17 = 0.60.
Highest state in North Central: Illinois and Michigan, tied at 0.9%.
Number of North Central states above the national median (0.95%) = 0 out of 12.

Headline: North Central Is America's Most Literate Region, at 0.7% Illiteracy

Evidence: In 1970 the North Central states averaged 0.70% illiteracy, only 0.60 times the national mean of 1.17%, and none of its 12 states was above the national median of 0.95%. Even its highest states, Illinois and Michigan (0.9%), were below that median, while the South averaged 1.74%, which is 1.04 points higher than North Central.

Recommendation: The minister should not spend literacy funds on North Central, and should instead target the South, especially Louisiana (2.8%) and Mississippi (2.4%).